

SCRM — Sensor Conflict Resolution Module

Parent Standard: Physical Reality Interpretation Layer (PRIL™)

Category: AI & Interpretation

Subcategory: Sensor Conflict Resolution

Type: Physical Interpretation Module

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Compatibility: OOF® Methodology OS™ · PRIL™ · CLIA® · RIS · INTEGROS® · SIMULOS®

Authority: OOF®

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Canonical Language: English (UCL™)

Canonical Definition

Sensor Conflict Resolution Module defines the methodological conditions under which autonomous systems detect, interpret, and resolve conflicting physical-world inputs from multiple sensing sources before generating physical action or embodied response.

A system satisfies SCRM only if:

- conflicting sensor inputs are recognized before execution
- conflict conditions are interpreted under defined resolution logic
- unresolved contradiction is treated as an execution restriction condition
- confidence degradation caused by sensor conflict is explicitly governed
- physical action is limited when sensor conflict remains unresolved or unreliable

A system that receives contradictory sensing inputs without governed conflict resolution does not satisfy SCRM.

Module Function

SCRM defines the conflict-resolution layer of physical interpretation.

It ensures that systems do not treat multimodal sensing as valid simply because multiple inputs exist.

The module applies wherever camera, lidar, radar, sonar, thermal, depth, mapping, or other physical sensing inputs may produce inconsistent, contradictory, or confidence-degrading interpretations of the same environment.

Minimum Implementation Framework (MIF)

Step 1 — Define Relevant Sensor Sources

The organization must define which sensing sources may contribute to physical interpretation.

Minimum requirement:

- relevant sensing channels are explicit
 - the interpretation role of each source is identifiable
 - undefined sensor dependence is excluded from valid execution logic
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Step 2 — Define Conflict Conditions

The system must define when sensing inputs are considered contradictory, unstable, or interpretation-breaking.

Minimum requirement:

- conflict thresholds are explicit
 - contradictory environmental readings are distinguishable
 - silent inconsistency is excluded from valid interpretation
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Step 3 — Define Resolution Logic

The system must define how conflict is interpreted and resolved.

Minimum requirement:

- resolution logic is explicit
 - preference, fallback, or restriction conditions are defined
 - hidden or arbitrary conflict handling is excluded
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Step 4 — Define Uncertainty and Restriction Conditions

The system must define when unresolved conflict slows, stops, restricts, or escalates execution.

Minimum requirement:

- degraded confidence is governed
 - unresolved contradiction can restrict action
 - continued execution under unresolved sensor conflict is not treated as valid
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Step 5 — Preserve Interpretation Integrity

The system must preserve physical interpretation integrity above speed, completion pressure, or optimistic assumption.

Minimum requirement:

- contradiction does not get silently normalized
 - execution convenience does not override conflict uncertainty
 - physical action remains subordinate to resolved interpretation conditions
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Step 6 — Restrict Invalid Execution

The system must not be treated as valid if physical execution proceeds while sensor conflict remains unresolved, hidden, or structurally ungoverned.

Minimum requirement:

- invalid execution conditions are identifiable
 - unresolved sensing contradiction blocks valid reliance
 - multiple inputs alone do not restore interpretation validity
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Use Case 1 — Autonomous Vehicle Sensor Disagreement

Scenario

An autonomous vehicle receives conflicting information from camera, lidar, and radar regarding obstacle presence, object distance, or movement pathway.

Application

SCRM is used to ensure:

- the conflict is recognized explicitly
 - contradiction thresholds are defined
 - unresolved disagreement restricts execution
 - physical action does not continue on ungoverned sensor assumption
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Result

The organization gains a stronger basis for preventing autonomous operation when multimodal sensing exists but physical interpretation remains structurally contradictory.

Use Case 2 — Industrial Robotics in Mixed Sensing Environment

Scenario

A robotic system operates in an environment where thermal, vision, depth, and positional sensing may produce inconsistent readings because of occlusion, heat distortion, reflective surfaces, or dynamic movement.

Application

SCRM is used to ensure:

- conflict conditions are explicitly governed
 - resolution logic is defined
 - degraded confidence restricts action
 - physical execution is not based on silent contradiction between sensing sources
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Result

The organization gains a structurally governed conflict-resolution layer instead of assuming that more sensors automatically produce valid physical understanding.

Canonical Closing Statement

If contradictory sensing is not resolved before action, physical execution is not structurally reliable.

Related Documents

→ [Physical Reality Interpretation Layer \(PRIL™\)](https://originopenfoundation.org/content/p/pril-scrm.html)

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Canonical Source

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